

ExaGuard AI

Enterprise observability and real-time governance for autonomous AI agents

ExaGuard AI is an inline security gateway between enterprise applications and large language models. It captures execution telemetry, scrubs sensitive personal data, intercepts adversarial prompt injections, and stores high-velocity logs directly inside Exasol Personal's in-memory columnar database.

By leveraging Exasol's high-speed analytical engine, ExaGuard lets security operations centers and compliance officers run sub-second queries across millions of agent interactions, track real-time LLM spend, enforce policy boundaries, and generate instant certifications for regulations such as the EU AI Act and SOC2 Type II.

Autonomous agents are outrunning enterprise oversight

Agent security

Autonomous agents execute tools without oversight; attackers exploit prompt injections and jailbreaks to extract database contents and hijack workflows.

Data privacy (PII)

Agents inadvertently ingest and expose sensitive customer data — SSNs, credit card numbers, medical records, API keys — in raw log files.

Operational costs

Multi-step agent reasoning loops can enter recursive deadlocks or token-exhaustion loops, burning thousands of dollars in LLM API fees within minutes.

MCP tool vulnerabilities

The Model Context Protocol lets agents invoke external server tools without parameter sanitization — opening the door to SQL injection and path traversal.

Traditional transactional databases can't aggregate this telemetry in real time — and manual audit prep takes weeks.

One inline gateway, one in-memory system of record

Inline interception gateway

Inspects prompt payloads, responses, and tool calls in under 40ms, automatically blocking prompt injections and role-override exploits.

Automated entity redaction

Scans every prompt and completion with regex and heuristic pattern scrubbers, replacing raw PII with safe masking tokens before persistence.

Dynamic policy engine

Enforces cost caps per transaction, maximum token boundaries, and recursion circuit breakers that terminate runaway agent loops immediately.

MCP security firewall

A dedicated JSON-RPC proxy that inspects MCP tool parameters in flight, preventing SQL injection and directory path traversal.

Behind all four sits Exasol's in-memory column store, running real-time aggregation for audit and compliance reporting.

Stopping attacks before they reach infrastructure

MCP SECURITY FIREWALL

First-of-its-kind JSON-RPC proxy. Intercepts tool requests between LLMs and MCP servers, inspecting inputs for malicious SQL syntax, shell commands, and directory traversal tokens.

Blocks the emerging attack surface. Prevents adversaries from using agentic tool calls as a path into internal enterprise infrastructure.

Runs in-line, not after the fact. Validation happens before a tool call reaches its destination server.

IN-DATABASE 3-SIGMA ANOMALY ENGINE

Native analytical SQL, in memory. Uses Exasol's window functions (AVG, STDDEV) to compute moving token and latency baselines across agent fleets.

Flags true outliers. Identifies transactions exceeding three standard deviations from the fleet baseline.

No external ML pipeline required. Detects zero-day exfiltration loops and anomalous behavior using the database itself.

From red-team simulation to one-click certification

ADVERSARIAL AI RED TEAMING SUITE

Seven structured attack vectors. DAN jailbreaks, indirect extraction, synthetic SSN leaks, token bombs, and privilege escalation, run against a target agent.

Quantifiable resilience ratings. Lets security teams benchmark model defenses before deploying agents into production.

NL-TO-SQL COMPLIANCE ASSISTANT

Semantic mapping engine. Translates human-readable audit questions directly into optimized Exasol columnar SQL.

One-click cryptographic certification. Generates verifiable scorecards against EU AI Act, SOC2 Type II, and HIPAA — cutting audit prep from weeks to seconds.

From agent fleet to sub-2ms analytical query

Enterprise Application Layer

Autonomous multi-agent fleet — FinTech, Health, DevOps, Support



ExaGuard Security & Governance Gateway

MCP firewall · Prompt-injection scrubber · PII redaction · Policy enforcement



Exasol Personal Columnar Database

AGENT_TRACES · SECURITY_ALERTS · POLICY_RULES · AUDIT_LOGS · 3-Sigma SQL engine



Security Operations Center UI

Telemetry waterfall · Red-team matrix · Compliance assistant · Certificate exporter

A production-grade layer on top of Exasol Personal

Layer	Technology	Function
Database & Analytics	Exasol Personal	Core platform — columnar telemetry storage, sub-second aggregation, in-database anomaly baselines
Database Driver	PyExasol	Native WebSocket connector for high-throughput batch imports and analytical queries
Backend API Gateway	FastAPI & Uvicorn	Asynchronous REST backend handling ingestion, MCP inspection routing, and production assets
Frontend Framework	React 18 & TypeScript	Component-driven, type-safe UI with reactive state management
Styling & Design	Tailwind CSS	Warm-cream enterprise SaaS theme with Lucide SVG iconography
Build & Bundling	Vite	Compiles optimized static bundles into the backend distribution layer
Deployment	Vercel	Serverless deployment for the production build and API routes

Four columnar tables, built for analytical speed

Table	Purpose & Exasol Acceleration
AGENT_TRACES	Primary columnar table recording every agent interaction — tokens, latency, cost, PII and injection flags. Columnar compression enables sub-second aggregation on numeric columns.
SECURITY_ALERTS	High-priority registry tracking mitigated injection attacks, PII exposures, unauthorized tool attempts, and policy violations.
POLICY_RULES	Configurable operational quotas, token ceilings, cost limits, and restricted system tool whitelists.
AUDIT_LOGS	Regulatory audit records capturing snapshot evaluations for EU AI Act, SOC2 Type II, and HIPAA compliance.

Each table's primary columns follow a consistent naming pattern (TRACE_ID, AGENT_NAME, SEVERITY, POLICY_ID ...) to keep cross-table joins analytics-friendly.

Six interactive modules, one SOC console

Executive SOC Overview

Real-time KPI cards, threat distribution, agent spend, 3-Sigma anomaly tables, live trace waterfalls.

Security & Guardrail Sandbox

Test custom prompt injections and synthetic PII leaks against the gateway in real time.

MCP Security Firewall

Inspector for MCP JSON-RPC parameters — blocks SQL injection and directory traversal.

AI Red Teaming & Stress-Tester

One-click battery of 7 adversarial vectors, scoring defense resilience per agent.

Exasol Speed Benchmark

Multi-dimensional aggregation runner comparing Exasol against traditional row-store estimates.

Compliance Auditor & Assistant

Scorecards for EU AI Act, SOC2, and HIPAA, plus a natural-language query bar.

How ExaGuard addresses each evaluation dimension

Criterion	How ExaGuard AI Addresses It
Effective use of Exasol Personal	Exasol is the computational engine, not a passive sink. High-velocity telemetry streams into column-store tables; analytical SQL executes multi-dimensional aggregation and in-database 3-Sigma anomaly scoring in under 2 milliseconds.
Innovation & problem impact	Tackles the top barrier to scaling autonomous agents — security, trust, and auditability — with a first-of-its-kind MCP JSON-RPC firewall and in-database statistical anomaly detection.
Technical excellence	Full-stack, production-grade build: type-safe React/TypeScript frontend, asynchronous FastAPI backend, native PyExasol integration, resilient fallback mode, containerized packaging.
Solution design & user experience	A clean, warm-cream enterprise aesthetic — white cards, slate typography, Lucide icons — with six deep interactive modules and no informal emoji.
Presentation & documentation	Structured pitch deck, a timed demo guide, Apache 2.0 license, a verified submission checklist, and a clean initial commit on GitHub.

Govern autonomous agents at the speed they run at — before a runaway loop becomes an incident report.

One inline gateway, one in-memory system of record, and instant compliance certification — built on Exasol Personal.

Thank you.